

FILE 0011744619 · AUSTRALIA · 2026 – 2028

Field *Pack*

A working reference for Australian agronomists evaluating SPANEX encapsulated microbial soil inoculants. Programme design, dose tables, application logistics, and field-trial evidence — without the calendars.



DOCUMENT SET

14 sheets

SKU RANGE

8 organisms

CFU PER CAP

10¹² activated

DAFF PERMIT

0011744619

A WORKING INDEX

What's *inside* this pack

Read in order or pick up at any tab. Marketing materials sit at the front for context only — the agronomic substance starts at sheet 04 and lives in the data.

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A NOTE BEFORE THE DATA

Read this *first*

SPANEX is an encapsulated soil microbial inoculant. It is **not** a fungicide, a fertiliser substitute, or a magic bullet. It is a delivery format for eight specific organisms with documented mechanisms of action and a peer-reviewed encapsulation patent.

This pack is structured the way an Australian agronomist would actually use it: organisms first, then how to apply them, then what we have evidence for, then what we are still building. The marketing-style summaries are at the back of the section dividers, not the front.

What you should expect from this pack

- Eight one-page product sheets — strain identity, mechanism, per-strain CFU, agronomic threshold, crop fit.
- An application guide that names water volumes, growth stages and re-entry intervals (nil, in our case).
- A tank-mix compatibility statement that lists what we are not compatible with — because that's the bit you actually need.
- A blank recommendation record sheet you can use in the paddock.
- The five Indian field trials we have on file, statistically re-framed — not the farmer ratings.
- Two open templates: our letter to the manufacturer asking for independent data, and our outreach to Australian universities.

What this pack does not contain

- Australian university trial data — we don't have it yet. The §IV section explains how we are commissioning it.
- FRAC / HRAC / IRAC codes — SPANEX is not a pesticide, so they don't apply.
- PHI / MRL data — biofertilisers carry no withholding period; this is stated where required.
- Yield-uplift superlatives. The numbers in §III are reproducible from the source FTR PDFs.

HONEST FRAMING

SPANEX has Indian field trial data and manufacturer CoAs. It does not yet have AU-specific independent trial data. We do not claim it does. We do recommend a paid pilot trial as the path to AU-specific data — terms on §20.

HOW TO USE THIS PACK

The pack is print-bound. Tear out §14 (the recommendation record sheet) and use it in the paddock. Keep §15–18 as your evidence reference. Hand §20 (the letter / outreach) to your research lead if you want to co-fund a trial.

We brought the data, not the calendars.

MEEM INTERNATIONAL PTY LTD
IMPORTER & AUSTRALIAN
DISTRIBUTOR
DAFF PERMIT 0011744619
MANUFACTURED BY SRT AGRO
SCIENCE PVT LTD
UNDER LICENCE OF ICAR-IISR
PATENT IN 361021/CHE/2013

SHEET 01 OF 8 – N-FIXER (AEROBIC / FREE-LIVING)

Azoto Caps

Azotobacter chroococcum NCIM 2632

Burns oxygen at extreme respiratory rates to protect nitrogenase, fixing atmospheric $N_2 \rightarrow NH_4^+$ in the bulk soil and rhizosphere. Carbon-limited, broad-host.

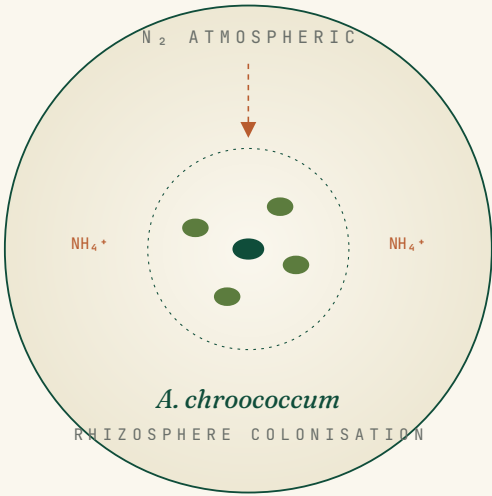
Aerobic

Free-living

Broad-host

Endospore-stable

BSL-1



CFU & DOSE	
CFU PER CAP (POST-ACTIVATION)	1 × 10 ¹² (peak; 7 × 10 ¹² observed in batch CoA 203)
CFU AS-SUPPLIED (IN-CAP)	min 5 × 10 ⁷ – activated by 6–8 hr soak
AU RATE	1 capsule per hectare per application
PER-STRAIN AGRONOMIC THRESHOLD	10 ¹¹ – 10 ¹² CFU / ha (LIT)
SAFETY FACTOR OVER LIT MINIMUM	1× – 10× depending on strain
APPS PER CYCLE (ANNUAL CEREAL)	1 – at sowing
RE-APPLICATION WINDOW	90 – 120 days controlled release

CROP FIT & EXCLUSIONS	
CROP FAMILY	PROGRAMME ROLE
Cereals – wheat, barley, oats	Lead N-fixer
Cotton	Lead N-fixer + PSB
Tree fruit · vine · berries	Lead N-fixer
Tomato · cucurbits · veg	Co-N (with Azoss)
Pasture (grass-only)	Co-N (with Azoss)
Legumes (any host)	Use Rhizo, not Azoto
Sugarcane	Use Aceto, not Azoto

COMPOSITION (PER CAP)

ACTIVE MICROBIAL CULTURE	5 % w/w
CARRIER – PHARMACEUTICAL TALC	94 % w/w · 0.150–0.212 mm
STABILISER – AGAR / ALGINATE 50:50	0.5 % w/w
RESIDUAL MOISTURE	≤ 1 % w/w
CAPSULE SHELL – HPMC VEGETAL	0.12 g · CAS 9004-65-3
TOTAL MASS	0.9 – 1.0 g + shell

STORAGE & HANDLING

TEMPERATURE TOLERANCE	5 °C – 50 °C (optimum 25–30 °C)
STORAGE	Cool / dry, ambient · < 25 °C
SHELF LIFE (COA-STAMPED)	16 months from MFG · 19 mo tested
COLD CHAIN	Not required
RE-ENTRY INTERVAL	Nil
WITHHOLDING PERIOD	Nil
REGULATORY CATEGORY	Biofertiliser · NOT APVMA

WHAT AZOTO CANNOT DO

Heavy synthetic urea suppresses nitrogenase activity – Azoto is most effective when partial-rate N is run, not full-rate. On low-OC sandy soils (<1 % SOC), carbon limitation reduces fixation; pair with humic acid input.

SHEET 02 OF 8 – N-FIXER (MICROAEROBIC / ASSOCIATIVE)

Azoss Caps

Azospirillum brasilense NCIM 5135

Hides in the microaerobic film 1 – 2 mm against grass-family roots. Lower energy intake, lower biomass; advantage is host-specific colonisation of grass rhizospheres including paddy, cane and pasture grasses.

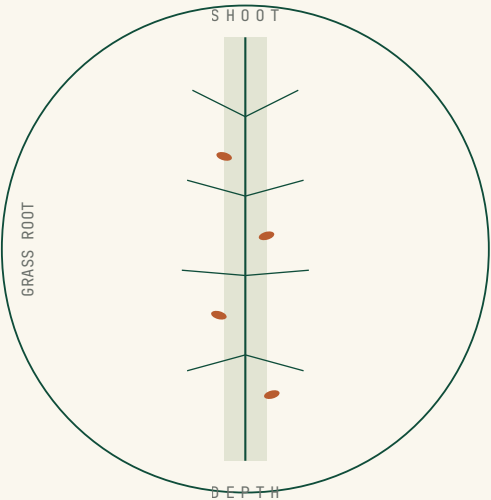
Microaerobic

Associative

Grass-family

Rhizosphere niche

BSL-1



CFU & DOSE

CFU PER CAP (POST-ACTIVATION)	1 × 10 ¹² peak
AU RATE	1 capsule per hectare per application
PER-STRAIN THRESHOLD	10 ¹¹ – 10 ¹² CFU / ha
APPS PER CYCLE	Cereals 1 (sowing); paddy 2 (transplant + tillering); pasture 3 / yr
RE-APPLICATION WINDOW	90 – 120 days SRT

CROP FIT & EXCLUSIONS

CROP FAMILY	PROGRAMME ROLE
Paddy (rice)	Lead N-fixer
Sorghum · maize · oats forage	Lead N-fixer
Pasture grass component	Lead – pair Rhizo (clover)
Cereals (wheat, barley)	Co-N (with Azoto)
Sugarcane	Co-N (with Aceto)
Tree fruit · vine · berries	Use Azoto, not Azoss
Legumes	Use Rhizo

MECHANISM – N-FIXATION, KG N/HA/YR	
CROP CONTEXT	RANGE (LIT)
Cereals · paddy	5 – 25
Mixed pasture grasses	5 – 20
Sorghum · maize	10 – 25
Plant-uptake basis (Y1)	30 – 60 % of above

STANDARD PARAMETERS	
CELL WEIGHT PER CAP	0.9 – 1.0 g + 0.12 g shell
SHELF LIFE	16 months CoA · 19 mo tested
STORAGE	Cool / dry · < 25 °C
COLD CHAIN	Not required
RE-ENTRY INTERVAL	Nil
REGULATORY	Biofertiliser · NOT APVMA

OPERATIONAL NOTE

Azoss is microaerobic – apply after first irrigation onto moist soil, not into dry seedbed. Better through fertigation than pre-sowing broadcast.

SHEET 03 OF 8 — N-FIXER (ENDOPHYTIC / SUGARCANE SIGNATURE)

Aceto Caps

Gluconacetobacter diazotrophicus NCIM 5348

Lives inside sugarcane stem and root tissues — host regulates oxygen, host supplies sucrose. Highest energy supply of the four N-fix lifestyles in the SPANEX range. The single most differentiated SKU.

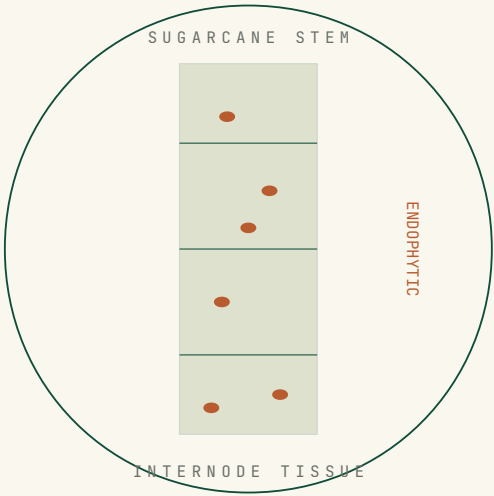
Sugarcane signature

Endophytic

Microaerobic (host-controlled)

Sucrose-fed

BSL-1



CFU & DOSE

CFU PER CAP (POST-ACTIVATION)	1 × 10 ¹² peak
AU RATE	1 capsule per hectare per application
APPS PER CANE CYCLE	2 — at planting + grand-growth (~5 mo)
PER-STRAIN THRESHOLD	10 ¹¹ — 10 ¹² CFU / ha

N-FIXATION POTENTIAL — KG N/HA/YR

CROP	RANGE	% OF CROP DEMAND
Sugarcane (signature host)	30 — 150	up to 60 %
Rice (paddy, secondary host)	10 — 40	15 — 30 %
Sorghum (rare host)	5 — 20	5 — 15 %

SHEET 03 · ACETO CAPS · *G. DIAZOTROPHICUS*

CROP FIT & EXCLUSIONS

CROP FAMILY	PROGRAMME ROLE
Sugarcane (Burdekin · Mackay · Bundaberg · Herbert)	Lead — irreplaceable
Paddy / rice	Co-N (with Azoss)
Sorghum (selected hosts)	Trial-only
Cereals · cotton · tree fruit · vine · pasture · veg	Out of host range

STORAGE & HANDLING

SHELF LIFE	16 months CoA
STORAGE	Cool / dry · < 25 °C
COLD CHAIN	Not required
RE-ENTRY INTERVAL	Nil

endophytic N-fix

SHEET 04 OF 8 — N-FIXER (SYMBIOTIC / LEGUME NODULE)

Rhizo Caps

Rhizobium leguminosarum MTCC 10096

Forms nodules in legume roots — host builds the dedicated organ, leghaemoglobin scavenges O₂. Highest-yielding biological N-fix system known. Group E biovar — clover, lucerne, leucaena, pea, lentil, faba.

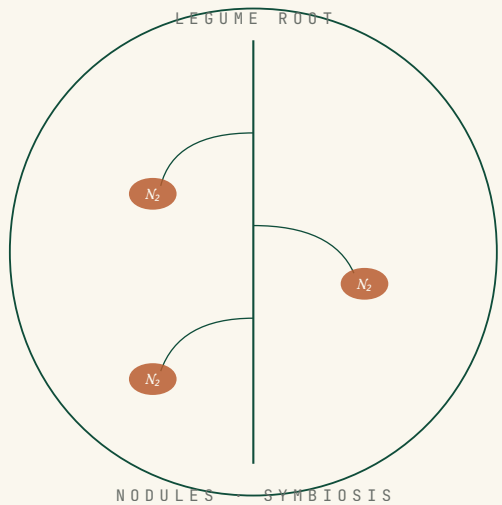
Group E biovar only

Symbiotic

Anaerobic in nodule

Plant-fed

BSL-1



CFU & DOSE

CFU PER CAP (POST-ACTIVATION)	1 × 10 ¹² peak
AU RATE	1 capsule per hectare per application
APPS — PULSE SEED-COAT	1 at sowing
APPS — PERENNIAL LEGUME / PASTURE	Establishment + biennial top-up

N-FIXATION POTENTIAL

SYSTEM	KG N/HA/YR
Lucerne stand	200 – 400
Leucaena (tropical)	150 – 300
Clover-based pasture	50 – 200
Pulse crop (pea, lentil, faba)	80 – 200

HONEST COMPETITIVE POSITION

SPANEX Rhizo Caps does not currently hold AIRG Green Tick Australian quality certification. For pulse seed-coat applications competing with EasyRhiz™ / NoduleN™ — concede. Position Rhizo for perennial legume programmes (pasture, lucerne, leucaena) where the SRT release window is the differentiator and competitors lose.

SHEET 05 OF 8 – SOLUBILISER (P)

PSB Plus Caps

Bacillus megaterium NCIM 2032

Secretes gluconic, citric, oxalic and lactic acids that chelate Ca^{2+} / Fe^{3+} / Al^{3+} holding bound P. Phytase / phosphatase release P from organic phytate. Endospore-forming – survives drought, freeze, UV.

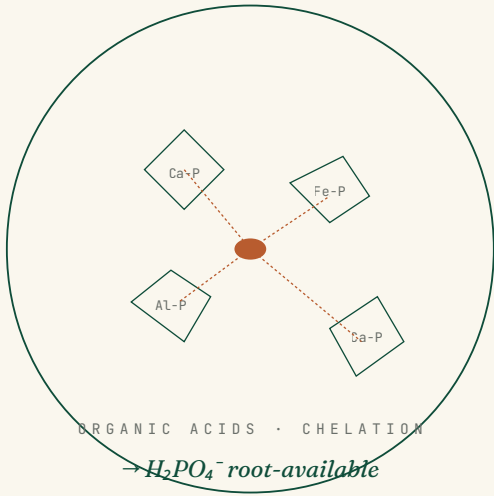
Solubiliser

Endospore

Universal crop fit

Acid-secreting

BSL-1



P RELEASED – KG/HA/SEASON

SOIL CLASS	P RELEASED
Vertosol (high Ca-P)	15 – 30
Ferrosol (high Fe-P · Al-P)	10 – 25
Calcarosol (high Ca-P)	10 – 25
Sandy / low-OC	5 – 15
Acidic / Sodosol	5 – 15

MODE OF ACTION

- Gluconic, citric, oxalic, lactic acid secretion → 0.5–1.0 unit pH drop in rhizosphere
- Chelation of Ca^{2+} / Fe^{3+} / Al^{3+} binding bound P
- Phytase + phosphatase enzymes – release organic P (phytate)
- Bonus: co-releases Zn / Fe in same chelation reaction

CROP FIT

APPLICATION	PRIORITY
P-locked Vertosols (cotton, leucaena, sorghum)	Critical
P-fixing Ferrosols (apples, banana, citrus, berries)	Critical
Ca-rich calcarosols (SA, Sunraysia)	High
All other crop-soil combinations	Standard

PROGRAMME ROLE

PSB Plus is the universal phosphate-economy SKU – included in every multi-SKU programme. On Vertosols and Ferrosols, double dose is justified by the soil P-fixation capacity.

SHEET 06 OF 8 — MOBILISER (K)

Potash Grow Caps

Bacillus decolorationis IARI-SL-18

Weathers K-bearing minerals (mica, feldspar, illite) via citric / oxalic / tartaric acid + extracellular polysaccharide chelation of K⁺ from clay lattices. Co-releases silicon — bonus cell-wall strength.

Mobiliser

Mineral weathering

EPS-producing

Silicon co-release

BSL-1



K RELEASED — KG/HA/SEASON

SOIL CLASS	K RELEASED
Vertisol / clay-rich	10 — 20
Granite-derived sandy soils	5 — 12
Mica / illite-rich (north QLD)	12 — 22
Calcarosol	5 — 12
Highly weathered (low K)	2 — 8

MODE OF ACTION

- Citric, oxalic, tartaric, gluconic acid weathering of K-feldspar, mica, illite
- Extracellular polysaccharide (EPS) chelation of K⁺ from clay interlayers
- Capsule layer accelerates mineral dissolution kinetics
- Co-releases Si⁴⁺ → cell-wall strength and drought tolerance bonus

CROP FIT (HIGHEST ROI)

CROP	PRIORITY
Cotton	Critical
Sugarcane	Critical
Banana · plantain	Critical
Potato · tubers	High
Tree fruit · citrus · vine	High
Pasture · cereals	Standard

STORAGE

SHELF LIFE	16 months CoA
COLD CHAIN	Not required
RE-ENTRY · WITHHOLDING	Nil · Nil

BEST ROI PER CAP

Highest measurable agronomic response on K-demanding crops (cotton, cane, banana, potato). For

SHEET 07 OF 8 — SOLUBILISER (ZN) + BIOCONTROL
PGPR

Zinc Grow Caps

Bacillus subtilis MTCC 8141

Two functions in one cap. Solubilises Zn-oxide / Zn-carbonate / Zn-phosphate via gluconic + 2-ketogluconic acid.
Plus secretes surfactin / iturin / fengycin lipopeptides — antifungal against *Fusarium* · *Rhizoctonia* · *Pythium* · *Botrytis*.

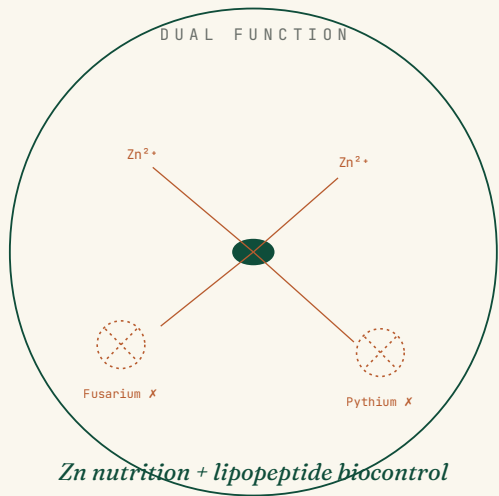
Zn solubiliser

Biocontrol

Lipopeptide-secreting

Endospore

BSL-1



ZN RELEASED + BIOCONTROL ACTIVITY

ACTIVITY	RANGE
Zn released — kg/ha/season	0.2 — 2.0
Crop Zn demand	0.3 — 0.5 kg/ha
Surfactin / iturin / fengycin secretion	Documented LIT
<i>Fusarium</i> suppression (in-vitro)	Yes
<i>Rhizoctonia</i> suppression	Yes
<i>Pythium</i> suppression	Yes
<i>Botrytis</i> suppression	Yes

HIGHEST-VALUE CONTEXTS

CONTEXT	PRIORITY
SA Eyre Peninsula · Mid-North · Mallee	Critical (Zn-locked Calcarosols)
WA wheatbelt — Zn-deficient zones	Critical
Sunraysia / Riverland (citrus, vine)	Critical (calcareous + Zn deficient)
NSW cotton (alkaline Vertosols)	High
Berries (Botrytis pressure)	High — biocontrol value
Cut flowers (Fusarium pressure)	High — biocontrol value

THE BIOCONTROL BONUS

For premium horticulture (berries, flowers, tomato, tree fruit), the lipopeptide biocontrol activity is often more economically significant than the Zn nutrition.

SHEET 08 OF 8 — ALL-IN-ONE CONSORTIUM SKU

NPK Grow Caps

3-strain consortium · combined N + P + K

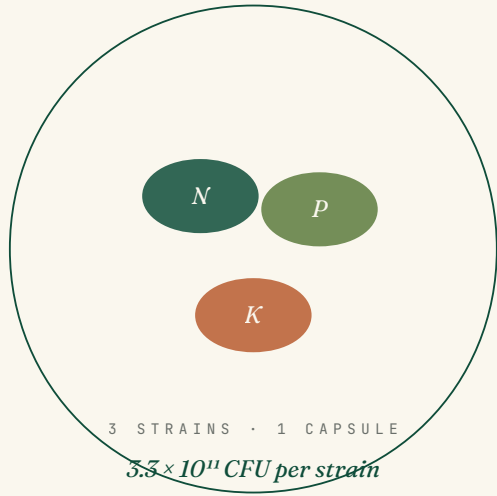
Single capsule containing three SPANEX strains pre-blended at 3.3×10^{11} CFU each. The simplification SKU — used as the entry / drop-in product when a multi-SKU programme is operationally too complex.

Entry SKU

Consortium

N + P + K combined

Apply 3 caps/ha



COMPOSITION (PER CONSORTIUM CAP)

N — AZOSPIRILLUM BRASILENSE NCIM 5135	3.3×10^{11} CFU
P — BACILLUS MEGATERIUM NCIM 2032	3.3×10^{11} CFU
K — BACILLUS DECOLORATIONIS IARI-SL-18	3.3×10^{11} CFU
TOTAL CFU PER CAP	1×10^{12} (split 3 ways)

WHY 3 CAPS/HA (NOT 1)

Each strain in NPK Grow is at 3.3×10^{11} CFU per cap — one third of the per-strain agronomic threshold. Apply **3 caps/ha** to bring each individual strain to 10^{12} CFU/ha threshold. Single-strain SPANEX SKUs require only 1 cap/ha.

WHEN TO USE NPK GROW VS FULL MULTI-SKU

DECISION DRIVER	CHOOSE
First-time customer · simplification needed	NPK Grow
Distributor with limited shelf SKUs	NPK Grow
Premium hort needing dialled-in dosing	Full multi-SKU
Sugarcane (needs Aceto)	NPK Grow + Aceto
Legumes (needs Rhizo)	NPK Grow + Rhizo
Zn-deficient soil	NPK Grow + Zinc Grow

CROP FIT

Universal entry product. Default for cereals, pasture, cotton, hort. Add specialty SKUs (Aceto, Rhizo, Zinc Grow) where crop biology requires it.

NPK GROW CANNOT REPLACE

Aceto (sugarcane endophyte), Rhizo (legume symbiont), or Zinc Grow (Zn solubilisation + biocontrol) — these strains are not in the NPK

— SHEET 12 — PROGRAMME BY CROP

Application by growth stage — water · timing · method

All rates are 1 capsule per hectare per SKU per application. Activate every cap by soaking 6–8 hr in 5 L clean chlorine-free water before dilution. Apply within 4 hours of dilution. Maintain 3–4 day gap from chemical fertilisers.

CROP FAMILY	GROWTH STAGE / TIMING	PROGRAMME – FULL MULTI-SKU	PROGRAMME – NPK GROW SHORTCUT	WATER L/HA	METHOD
Cereals Wheat · barley · oats	At sowing – single dose	Azoto 1 + Azoss 1 + PSB 1 + Potash 1 + Zinc 1 = 5 caps	NPK Grow 3 + Zinc Grow 1 = 4 caps	500	Seed slurry · in-furrow
Sorghum / maize / oats forage	Sowing + V6 booster	5 caps × 2 apps = 10 caps/cycle	4 caps × 2 = 8 caps	500	Fertigation · drench
Paddy (rice)	Transplant + tillering (2 apps)	Azoto 1 + Azoss 1 + PSB 1 + Potash 1 = 4 caps × 2	NPK Grow 3 + Azoss 1 × 2 apps = 8	500	Soil drench
Cotton	Sowing + squaring (~90 d)	5 caps × 2 = 10 caps	4 caps × 2 = 8 caps	500	In-furrow + side-dress
Sugarcane	Planting + grand-growth (~5 mo)	Aceto 1 + PSB 1 + Potash 1 = 3 caps × 2 = 6	Aceto 1 + NPK Grow 3 × 2 = 8	500	Setts + fertigation
Pasture – Y1 establishment	Spring + summer + autumn (3 apps)	Rhizo 1 + Azoss 1 + PSB 1 + Potash 1 = 4 × 3 = 12	Rhizo 1 + NPK Grow 3 × 3 = 12	500	Boom spray on moist soil
Pasture – Y2+ maintenance	1–2 apps / yr (start of growth season)	4 caps × 1–2 = 4–8 caps	4 caps × 1–2 = 4–8	500	Boom · oversow with seed
Lucerne (perennial)	Spring post-cut + autumn post-cut (2 apps)	Rhizo 1 + PSB 1 + Potash 1 = 3 × 2 = 6	Rhizo 1 + NPK Grow 3 × 2 = 8	500	Drench · fertigation
Leucaena (AACO)	Establishment + biennial top-up	3 caps × 2 = 6 caps	4 caps × 2 = 8 caps	500	Seed slurry establishment + boom for top-up
Pulses – chickpea, lentil, faba	Seed-coat at sowing (single)	Rhizo 1 + PSB 1 + Potash 1 = 3 caps	Rhizo 1 + NPK Grow 3 = 4 caps	500	Seed slurry
Tree fruit · vine	Bud-break + summer + post-harvest (3 apps)	Azoto 1 + PSB 1 + Potash 1 + Zinc 1 = 4 × 3 = 12	NPK Grow 3 + Zinc 1 × 3 = 12	500	Drip / fertigation
Berries (perennial)	Pre-flower + post-harvest (± summer)	5 caps × 2–3 = 10–15 caps	4 caps × 2–3 = 8–12	500	Drip / fertigation
Tomato · veg	Transplant + fruit-set (1–2 apps)	5 caps × 1–2 = 5–10 caps	4 caps × 1–2 = 4–8	500	Drench · drip
Cut flowers	Pre-flower + post-flower (3 apps)	5 caps × 3 = 15 caps	4 caps × 3 = 12	500	Glasshouse drip · fertigation

ACTIVATION PROCEDURE (EVERY CAP, EVERY TIME)

1 – Soak capsule(s) in 5 L lukewarm clean chlorine-free water · 6–8 hours (overnight). 2 – Stir / shake intermittently 3 – Microbe count multiplies ~10x during

CONDITIONS TO AVOID

Direct sunlight on mixed solution. Hot midday application. Chlorinated municipal water. Tank-mix with bactericides, copper-based fungicides, hydrogen

SHEET 13 – COMPATIBILITY STATEMENT

Tank-mix compatibility — the part you actually need

STATEMENT HONESTY

This sheet lists what we have **data** on. Many farm-chemistry combinations have not been bench-tested with SPANEX. Treat anything not on this list as "test in a small jar before tank-loading". Manufacturer-supplied compatibility data is being expanded — see §IV letter to SRT.

DO NOT MIX — CONFIRMED INCOMPATIBLE

CLASS	REASON
Bactericides (any)	Kills inoculum directly
Copper-based fungicides — Cu(OH) ₂ · CuSO ₄ · Cu-oxychloride · Bordeaux	Cu ²⁺ ion toxic to <i>Bacillus</i> spores
Chlorinated water (municipal · pool · superchlorinated)	Free Cl ₂ disinfects inoculum
Hydrogen peroxide / peracetic	Oxidises microbial cells
Quaternary ammonium sanitisers	Membrane disruption
Hot solution (> 40 °C)	Heat kills vegetative cells

SPACE BY 3–4 DAYS — SYNTHETIC FERTILISER TIMING

INPUT	SPACING
Urea (full rate)	3–4 days before/after
DAP / MAP	3–4 days
MOP / SOP	3–4 days
Anhydrous ammonia	7+ days minimum

LIKELY COMPATIBLE — PENDING FORMAL BENCH DATA

CLASS	NOTES
Most herbicides post-application	Apply SPANEX after herbicide is rainfast
Most insecticides (non-bactericidal)	Test small batch first
Humic / fulvic acid products	✓ Synergistic — supports establishment
Seaweed / kelp extracts	✓ Compatible
Compost extracts / teas	✓ Compatible
Mycorrhizal inoculants	✓ Compatible (often co-applied)
Foliar micronutrient chelates (Fe / Mn / B)	Test compatibility for jar test
Sulfur (elemental, dustable)	Apply 2–3 days apart

JAR-TEST PROTOCOL (5 MIN IN THE SHED)

1. Mix the proposed tank-mix at the same dilution as paddock rate, in 250 mL clean water.
2. Add the activated SPANEX solution to the jar.
3. Cap, invert 5 ×, leave 30 min.
4. Check for: flocculation, clouding, layer separation, gas-off (foaming), heat.
5. If clear or only mild clouding — proceed at small batch first.

DO NOT MIX list

FIELD-SIDE WORKSHEET · EDITION 01

Form RR-01

Recommendation Record

Date: _____
Sheet ____ of ____

A · FARM & PADDOCK

FARMER / BUSINESS NAME

CONTACT / PHONE

REGION / DISTRICT

PROPERTY NAME

PADDOCK ID

PADDOCK AREA (HA)

SOIL ORDER (ASC)

LAST SOIL TEST (DATE)

B · CROP & STATE

CROP / CULTIVAR

GROWTH STAGE

SOWING / TRANSPLANT DATE

PEST / DISEASE PRESSURE OBSERVED

NUTRIENT DEFICIENCY SYMPTOMS

C · SPANEX RECOMMENDATION

☐ Azoto Caps ☐ Azoss Caps ☐ Aceto Caps ☐ Rhizo Caps ☐ PSB Plus Caps

☐ Potash Grow Caps ☐ Zinc Grow Caps ☐ NPK Grow Caps

TOTAL CAPS RECOMMENDED

APPS PER CYCLE

FIRST APP PLANNED DATE

APPLICATION METHOD

CARRIER WATER (L/HA)

D · TANK-MIX · TIMING CONSTRAINTS

OTHER INPUTS IN 7 DAYS BEFORE / AFTER

JAR-TEST RESULT (IF PERFORMED)

F • SIGN-OFF

AGRONOMIST SIGNATURE

FARMER / MANAGER SIGNATURE

SPANEX REP SIGNATURE

Form RR-01 is a working field tool. Retain for paddock records and for SPANEX feedback if the paddock becomes a pilot reference site. SPANEX may use anonymised data from this sheet for product development and case study purposes — opt out by ticking here ☐.

SHEET 15 — FIVE FTRS AT A GLANCE

Indian field-trial summary — statistical reframing

The original FTRs are written as farmer satisfaction reports. This sheet re-states the evidence in the form an agronomist can audit.

FTR	FARMER · LOCATION	CROP · CULTIVAR	PERIOD · AREA	DESIGN	RESULT	N	EVIDENCE CLASS
#1 Tomato	Mr. Prakhar Varu · Sirsa, Bhilai-3, Durg, Chhattisgarh	Tomato (Saaho)	Nov 2022 – Mar 2023 · 2.5 ac	Paired Trial vs Non-Trial · full fertiligation logs both	+8–9 % yield · –19–20 % cost	1 × 1	FIELD-IN · MED
#2 Grapes	Mr. Mahaveer Patil · Takali, Miraj, Sangli, Maharashtra	Grape (Thomson)	Period not stated · 1 ac	Paired comparison · visual + farmer report	+20 % yield · 0 disease	1 × 1	FIELD-IN · MED-LOW
#3 Chickpea	Mr. Nutan Verma · Khangar Patt, Berla, Chhattisgarh	Chickpea	Nov 2023 – Mar 2024 · 4 ac	Single demo, no control	3,025 kg/ha	1 demo	FIELD-IN · LOW
#4 Wheat	Mr. Nutan Verma · same farm	Wheat	Nov 2024 – Mar 2025 · 3 ac	Single demo, no control · repeat customer	2,594 kg/ha	1 demo	FIELD-IN · LOW
#5 Tomato (dup)	Same as #1	Same	Same	Duplicate PDF on file	– see #1	–	–

TOMATO — PAIRED FIELD

+8–9 %

yield over Non-Trial @ 78.43 vs 72.06 MT/
acre

TOMATO — PROGRAMME COST

–19–20 %

₹80,120 vs ₹98,711 per acre

GRAPES — DISEASE INCIDENCE

0

bacterial / fungal infections in trial; vs
patches of damage in non-trial

STATISTICAL DISCLOSURES THE AGRONOMIST WILL ASK ABOUT

n: One paired comparison per FTR. No replication within or across seasons. **Confidence interval:** Cannot be calculated from n=1 paired data — these are point estimates, not statistically significant differences. **Programme isolation:** All trials use combined SRT programmes (SPANEX bio-capsules + PROM phosphorus manure + mycorrhiza + biocontrol kits + foliar micronutrients) — SPANEX in isolation has not been tested. **Soil class:** Indian Black Soil (Vertisol-equivalent) only.

Australian transferability: Indicative — pending paid-pilot trial confirmation.

SHEET 16 – FTR #1 DEEP DIVE

Tomato controlled trial — full data sheet

FARMER	Mr. Prakhar Varu, DOB 28 March 1994
ADDRESS	Kumhari, Durg, Chhattisgarh, India
PROPERTY	30 acres total · 2.5 acres demonstration
CROP · CULTIVAR	Tomato — Saaho variety
SOIL CLASS	Black Soil (Vertisol-equivalent)
PREVIOUS CROP	Pumpkin
CROPPING METHOD	Transplantation
IRRIGATION	Drip
TRIAL START · END	20 Nov 2022 — 13 Feb 2023
DECLARATION SIGNED	29 March 2023 by farmer
DESIGN	Paired Trial Field vs Non-Trial Field with full fertigation logs both

PROGRAMME – SRT TRIAL FIELD

COMPONENT	ACTIVE	QUANTITY / HA	MODE
IBP Kit	<i>Trichoderma viride</i> · <i>Pseudomonas fluorescens</i> · <i>Paecilomyces lilacinus</i>	1 kit	Soil drench
Mycorrhiza Granule	Mycorrhiza fungi	7.5 kg	Basal
Mycorrhiza NP	Mycorrhiza nano-powder	200 g	Soil
Amavasya Kit	5-bacteria pest/fungal control	2 kits	Foliar
Poornima Kit	NPK Consortia + Zinc Grow + nano-powders + mycorrhiza	0.5 kit	Drip
NPK Grow capsules	NPK Consortia (SPANEX)	16 caps	Soil
Azoto + PSB Plus + Potash Grow	SPANEX bio-capsules	2 + 2 + 2 caps	Soil
ALP + Sonha-Bihan	Chelated multi-micronutrient + PGR	2.5 L + 3 kg	Foliar
Humigrow + Cropforce	Humic acid + bio-zhyme	900 g + 300 g	Soil

DOCUMENTED OUTCOMES

METRIC	NON-TRIAL FIELD	SRT TRIAL FIELD	Δ
Yield	72.06 MT / acre	78.43 MT / acre	+8-9 %
Total programme cost	₹98,711 / acre	₹80,120 / acre	-19-20 %
Farmer rating	—	GOOD	Signed declaration
Best-feature feedback	"Easy to use, transport, and fast result"		

DEFENSIBLE CUSTOMER-FACING CLAIM FROM FTR #1

"In a 2022-23 controlled paired-paddock comparison on 2.5 acres of tomato (Saaho variety) on Indian Black Soil at Mr. Prakhar Varu's farm in Sirsa, Bhilai-3, Chhattisgarh, the integrated SRT/SPANEX programme delivered an 8-9 % yield increase and 19-20 % programme cost reduction vs the farmer's conventional programme. Full fertigation logs available."

SPANEX-isolated efficacy not separable from the integrated programme. Statistical replication and Australian-conditions trial pending.

SHEET 17 – FTR #2

Grapes — Maharashtra

FARMER	Mr. Mahaveer Patil
ADDRESS	Takali, Miraj, Sangli, Maharashtra
CROP · CULTIVAR	Grape (Thomson)
TRIAL AREA	1 acre (rest = farmer regular practice)
INSPECTING AUTHORITY	Mr. Shirish Taunk, MD, SRT Agro Science

PRE-TRIAL FIELD CONDITION

- Poor vegetative growth — plants not producing new branches or healthy leaves
- Disease & pest pressure — fungal + bacterial infections + nematode attacks
- Soil health — low microbial activity, poor nutrient availability
- Reduced yield in previous seasons

PROGRAMME APPLIED

1. IBP Kit (Trichoderma + Pseudomonas + Paecilomyces) — drip drench, repeated 15 d due to disease pressure
2. NPK Grow 2 caps + Humigrow 500 mL + Mycorrhiza Nano 100 g — drip, 1 wk after IBP
3. NPK Grow 2 caps + Humigrow 500 mL — drip, 1 mo later
4. ALP 250 mL + Sonha-Bihan 250 g — foliar at flowering
5. ALP 250 mL + Sonha-Bihan 250 g — drip at fruit-set

DOCUMENTED OUTCOMES

- +20 % farmer-reported yield increase
- 0 bacterial / fungal infections in trial field
- Patches of bacterial damage in non-trial field
- Visible difference photographed at multiple growth stages

SHEET 18 – HONEST CAVEATS

What the FTRs *do not* prove

REPRODUCIBILITY LIMITS

None of the 5 FTRs are statistically replicated. n = 1 farmer per trial. The two paired comparisons (Tomato, Grapes) are point estimates, not significance-tested differences. They are demonstrations, not proofs.

PROGRAMME ISOLATION

All trials apply combined SRT programmes — SPANEX bio-capsules plus PROM, mycorrhiza, biocontrol kits, foliar sprays, humic / fulvic acid. SPANEX-isolated efficacy in isolation is not measured. The 8–9 % tomato yield uplift is attributable to the integrated programme, not to SPANEX alone.

GEOGRAPHIC TRANSFERABILITY

All trials on Indian Black Soil (Vertisol-equivalent) in Chhattisgarh and Maharashtra. Australian regional / soil / climate / cultivar transferability is plausible but unverified.

WHAT THE DISEASE-SUPPRESSION IN FTR #2 ACTUALLY MEANS

The Grapes trial documents zero bacterial / fungal infections in SPANEX-treated. However the IBP Kit (containing *Trichoderma* · *Pseudomonas* · *Paecilomyces*) is the dedicated biocontrol component of the programme — these strains, not the SPANEX bio-fertilisers, are likely doing most of the disease suppression work.

PATH TO AU VALIDATION

See §IV (page 22) — university outreach pack template + paid pilot trial offer. Recommended first AU pilot: a single-region single-crop paired-paddock trial mirroring the Indian Tomato design, in Vertisol cotton country (Liverpool Plains or Darling Downs) or in AACO Vertisol leucaena country.

Meem International Pty Ltd
ABN – Australian distributor
Date: _____

SRT Agro Science Pvt. Ltd.
Village Funda, Tah: Patan
District Durg, Chhattisgarh
491 111, India
Attn: Mr. Shirish Taunk, Managing Director

Re: Request for additional product data — Australian agronomist channel

Dear Mr. Taunk,

As Australian distributor of the SPANEX biocapsule range, we are presenting your products to professional agronomy teams at AACO, Costa Group, Sugar Research Australia and major distributors. To meet their technical-review thresholds we require additional manufacturer-supplied data items, which we believe SRT already holds or can prepare quickly.

SPECIFICALLY REQUESTED

- 1. Updated Certificates of Analysis** for the HPMC-shell version of all eight SKUs (the CoA on file dated 7 Dec 2023 references the legacy Hard Gelatin shell). One CoA per SKU for the most recent manufacturing batch.
- 2. Per-strain CFU breakdown** for NPK Grow Caps — confirming the assumed equal-split (3.3×10^{11} per strain) or stating the actual split.
- 3. Extended stability data** at 18, 22 and 30 months under ambient ($< 25^\circ\text{C}$) storage to support a longer Australian shelf-life claim than the current 16-month CoA stamp.
- 4. SRT release-rate curve** documenting the controlled microbial release profile under standard soil moisture / temperature conditions.
- 5. Tank-mix bench compatibility data** for SPANEX $\times$ the following Australian-relevant chemistry: glyphosate, paraquat, bifenthrin, chlorpyrifos, mancozeb, tebuconazole, propiconazole, and calcium nitrate.
- 6. Independent third-party trial reports** — any ICAR, IISR or Indian state-agricultural-university trials of SPANEX (separate from on-farm SRT demonstration FTRs) that can be cited as independent evidence.
- 7. Comprehensive FTR archive** beyond the five we currently hold (wheat, chickpea, tomato, grapes) — particularly any sugarcane, cotton, paddy, lucerne or pasture FTRs available.

In return we will share back: customer-facing technical sheets, the field demonstration evidence pack we have built around your FTRs, and feedback from the Australian agronomy reviewers.

Please confirm timelines for each item by reply.

With sincere regards,

For Meem International Pty Ltd
Australian Distributor • DAFF Permit 0011744619

SHEET 20 — CO-FUNDED RESEARCH OUTREACH

University outreach pack — co-funded pilot proposal

For research leads at Murdoch · UQ · La Trobe · Sydney · Adelaide · CSIRO. SPANEX is offering co-funded paid-pilot AU field trials in exchange for independently authored data and publication rights.

WHAT WE OFFER

- Free product (full SKU range) at trial dose
- Cash co-funding (\$25–\$80K depending on scope)
- R&D Tax Incentive eligibility (43.5 % refundable for SME co-funding portion)
- GRDC / AgriFutures / MLA Donor Co. / CSIRO Innovation Connections grant alignment
- Anonymised farmer-cooperator network (Mr. Verma, Mr. Varu, Mr. Patil — Indian reference + AU pilot prospects)
- Publication rights to all data — university authored
- SPANEX agreement to publish negative results unedited

WHAT WE ASK

- One paired-paddock trial in priority region (see §III for region selection)
- Full statistical reporting including CI and $\alpha=0.05$
- Quarterly progress data shares
- Right of first review on draft publications (5 working days)
- If trial succeeds — co-authored case study for SPANEX customer-facing material

PRIORITY TRIAL REGIONS (RANK ORDER)

REGION	CROP	SOIL ORDER
QLD/NT Mitchell Grass Downs (AACO core)	Leucaena	Vertosol
QLD Burdekin / Mackay	Sugarcane	Vertosol / Dermosol
NSW / QLD Cotton Belt	Cotton	Vertosol
VIC Gippsland · Tas Northern	Dairy pasture	Dermosol / Ferrosol
SA Eyre + Mid-North	Wheat	Calcarosol
Sunraysia · Riverland	Wine grape	Calcarosol

TRIAL DESIGNS SPANEX IS OPEN TO

1. Paired-paddock — Trial vs farmer's existing programme (mirrors Indian FTR #1 design)
2. Replicated split-plot — 3 reps × 2 seasons, single farm
3. Multi-site — same crop across 3+ farms in one region
4. Dose-response — 0, 0.5, 1, 2, 4 caps/ha across 4–6 plots; generates the missing efficacy curve

WHAT WE WILL NOT DO

Constrain publication. Edit unfavourable findings. Pressure for positive-only reporting. Withhold negative results. Misrepresent trial methodology in marketing.

Meem International Pty Ltd
Email: bsn-group@outlook.com
DAFF Permit 0011744619

Research lead · institution · date